Energy Policy Review: Orangeland 2024

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Executive summary

This review is the fourth in-depth assessment of Orangeland's energy policy conducted by the Global Economic Cooperation and Development (GECD) Secretariat, following earlier examinations published in 2008, 2014 and 2019. It was prepared by an examination team drawn from four member delegations and the GECD Directorate for Energy and Environment, and is based on a review mission to Valencourt and the northern industrial corridor held over eleven days in March 2024. The team met representatives of the Ministry of Energy and Resource Security, the Orangeland Energy Regulatory Authority, the National Grid Operator, three regional distribution companies, industry federations, consumer associations and academic institutions. Written responses to a standard questionnaire, submitted by the Ministry in January 2024, provided the statistical foundation for the analysis. The draft findings were discussed with the authorities in June 2024 and the final text was approved by the GECD Standing Group on the Energy Market in September 2024.

Orangeland has made measurable progress since the 2019 review. Total primary energy supply fell by 4.7 per cent between 2019 and 2023 despite cumulative real output growth of 6.2 per cent over the same period, indicating a partial decoupling of energy use from economic activity. The share of low-carbon sources in the generation mix rose from 31 per cent to 44 per cent, driven mainly by the commissioning of the Vantry Sound offshore wind cluster and the second phase of the Kelloway solar programme. Greenhouse gas emissions from the energy sector declined by 12.8 per cent over the same interval, placing Orangeland close to, though not fully on, the trajectory implied by its 2035 statutory target.

Nevertheless, the review identifies three structural weaknesses that constrain further progress. First, energy intensity remains high relative to the GECD average, reflecting the continued weight of primary metals, cement and paper in the industrial base, together with a building stock of which an estimated 58 per cent predates the introduction of thermal standards in 1986. Second, the price signal for carbon remains weak and fragmented across sectors, with wide dispersion in effective rates and a large volume of free allocation. Third, permitting timelines for grid reinforcement average 61 months from application to authorisation, against a GECD median of 34 months, and now constitute the binding constraint on renewable deployment rather than project finance or supply-chain capacity.

The review's central message is that Orangeland's decarbonisation strategy is well designed at the level of targets but under-instrumented at the level of delivery. The 2022 Climate Framework Act established a credible governance architecture, including an independent advisory council and a five-yearly carbon budgeting cycle, yet the sectoral instruments that must deliver those budgets have not been recalibrated since 2020. The Secretariat encourages the authorities to treat the 2025 budget review as the occasion for a comprehensive re-examination of carbon pricing, network regulation and demand-side policy, rather than as a technical update of existing measures.

Energy supply and demand

Total primary energy supply in Orangeland stood at 87.3 million tonnes of oil equivalent in 2023, with natural gas the largest single fuel, followed by coal and nuclear; the shares by fuel for 2022 and 2023 are set out in Table 2.3 below. Domestic production covers approximately 41 per cent of supply, with the remainder imported, principally through the Marren Terminal and two cross-border pipelines. Import dependency has fallen by seven percentage points since 2019, largely because of reduced coal imports rather than growth in indigenous production. Gas storage capacity, at 4.1 billion cubic metres, is equivalent to 38 days of average winter consumption, which the review team considers adequate under central assumptions but thin under a combined cold-snap and supply-interruption scenario.

Final consumption reached 61.9 million tonnes of oil equivalent in 2023, distributed across industry (38 per cent), transport (27 per cent), residential (21 per cent) and services and agriculture (14 per cent). Industrial demand has been broadly flat in absolute terms since 2016 while industrial value added has grown, implying an intensity improvement of roughly 1.4 per cent per year. Transport demand, by contrast, rose by 3.2 per cent between 2021 and 2023 as travel patterns normalised, and the vehicle fleet remains dominated by internal combustion engines, with battery models accounting for 11.6 per cent of new registrations in 2023. Residential demand is highly weather-sensitive; the Secretariat's normalisation, which adjusts to a twenty-year heating degree day average, raises measured 2023 residential consumption by 2.9 per cent relative to the unadjusted figure reported by the Ministry.

Comparative intensity indicators place Orangeland toward the upper end of the GECD distribution. Orangeland remains the fourth most energy-intensive economy among GECD members, at 0.16 tonnes of oil equivalent per thousand units of GDP. This represents an improvement from the 2019 review, when the country ranked second on the same measure, but the pace of convergence has slowed since 2021. Decomposition analysis attributes about 62 per cent of the cumulative intensity gain since 2010 to structural shifts between sectors and about 38 per cent to efficiency improvements within them, a balance that suggests the scope for further structural gains may be narrowing. The review team notes that intensity comparisons of this kind are sensitive to exchange rate conversion and to the treatment of self-generation by industrial sites, and that the Ministry's own series, which uses a different deflator base year, yields a figure roughly two per cent lower.

Network infrastructure has become the principal physical constraint on the supply system. The transmission network comprises 18,400 kilometres of high-voltage lines, of which 22 per cent were commissioned before 1975 and are approaching the end of their design life. Curtailment of wind generation in the northern zones reached 6.8 per cent of available output in 2023, up from 2.1 per cent in 2020, and the connection queue for projects awaiting a firm offer stood at 31 gigawatts at the end of the review period. Interconnection capacity with neighbouring systems is equivalent to 9 per cent of peak demand, below the 15 per cent indicative benchmark adopted by the GECD Electricity Committee in 2021. The Secretariat considers that without a step change in the pace of reinforcement, curtailment could exceed 15 per cent by 2030 under the authorities' own central deployment scenario.

Table 2.3 Share of primary energy supply by fuel (%)

Fuel	2022	2023
Coal	31.2	27.4
Natural gas	26.8	28.1
Nuclear	24.1	26.0
Renewables	9.4	10.8
Oil	8.5	7.7

Decarbonisation policy

Orangeland's decarbonisation architecture rests on three pillars: the 2022 Climate Framework Act, which sets a statutory target of a 55 per cent reduction in economy-wide emissions by 2035 against a 2005 baseline; the national emissions trading scheme, introduced in 2016 and covering power generation and energy-intensive industry; and a set of sectoral instruments comprising vehicle standards, appliance regulation and building codes. The Framework Act establishes an independent Climate Advisory Council whose annual progress report must receive a formal government response within ninety days, an arrangement the review team regards as good practice. Carbon budgets are set five years in advance on a rolling basis, and the second budget period runs from 2026 to 2030.

The effectiveness of the trading scheme is limited by the level and dispersion of the price it produces. The carbon price averaged 21.4 currency units per tonne in 2023, well below the GECD median of 48.9. Free allocation covered 71 per cent of verified emissions in the industrial sectors, and a price containment mechanism introduced in 2019 releases additional allowances whenever the twelve-month moving average exceeds 30 units, effectively capping the signal. The scheme's coverage extends to approximately 46 per cent of national emissions, leaving transport, buildings, agriculture and waste outside its scope. When explicit and implicit pricing are combined, the Secretariat's effective carbon rate calculation yields an economy-wide average of 34 units per tonne, but with a range from 4 units in agriculture to 96 units in road transport, where fuel excise dominates.

Outside the trading scheme, the principal instruments are the Energy Efficiency Obligation placed on retail suppliers and the Warm Homes renovation grant. The obligation requires suppliers to deliver certified savings equivalent to 1.1 per cent of prior-year sales, a rate unchanged since 2018 despite the rising marginal cost of the remaining measures. Evaluation evidence commissioned by the Ministry in 2023 estimates an additionality factor of 0.63, implying that just under two-fifths of certified savings would have occurred in the absence of the obligation; the review team notes that this estimate rests on a matched-comparison design with a modest sample of 2,400 dwellings and should be treated as indicative rather than definitive. The Warm Homes grant has supported deep renovation of 84,000 dwellings since 2021, against a programme ambition of 250,000 by 2030, and uptake is concentrated among owner-occupiers in the upper half of the income distribution.

Innovation and industrial transition policy has expanded substantially. The Orangeland Clean Industry Fund, capitalised at 2.6 billion units over the period 2023 to 2030, supports electrification of process heat, hydrogen demonstration and carbon capture at two cement sites. The review team welcomes the fund's competitive allocation design but observes that its contract-for-difference instrument references a fixed strike rather than the prevailing allowance price, which weakens the interaction between the subsidy and the trading scheme. In the power sector, the auction framework for renewables has delivered falling clearing prices in five consecutive rounds, though the 2023 round was undersubscribed for the first time, with only 78 per cent of the offered volume awarded. The authorities attribute this to grid connection uncertainty rather than to the auction parameters themselves, an interpretation the Secretariat finds broadly consistent with the connection queue data.

Recommendations

On carbon pricing, the Secretariat encourages the authorities to strengthen the price signal and broaden its reach. The review recommends extending the emissions trading scheme to the buildings sector by 2028. It further recommends phasing out the price containment mechanism at the end of the current budget period, replacing it with a rising price floor indexed to the trajectory implied by the statutory target, and reducing free allocation on a published linear schedule so that industrial installations face at least half of their allowance requirement at auction by 2032. Revenue from expanded auctioning should be recycled in a manner that protects lower-income households, and the review team notes that a per-capita dividend would offset the average distributional impact for the bottom three deciles while preserving the marginal incentive to abate.

On networks and permitting, the review recommends that the authorities consolidate the current four-stage consent process for transmission reinforcement into a single procedure with a statutory eighteen-month determination deadline, supported by additional technical capacity in the regional planning authorities. Anticipatory investment should be permitted where the regulator judges that the option value of building ahead of firm demand exceeds the risk of stranded capital, and the current requirement for demonstrated committed load before capital approval should be relaxed accordingly. The review further recommends that connection queue management move to a milestone-based framework in which projects failing to meet defined development gates lose their queue position, and that interconnection capacity be raised toward the indicative benchmark through the two projects already at feasibility stage.

On demand-side policy, the Secretariat recommends recalibrating the Energy Efficiency Obligation to a rising target rate, beginning at 1.4 per cent from 2026, with a ring-fenced sub-target for dwellings in the lowest two income quintiles to address the uptake pattern identified in this review. Building codes should be tightened at the next scheduled revision to require near-zero operational demand in new construction from 2027, and a minimum performance standard should be introduced for the rental sector on a phased basis. The review also recommends that the Ministry publish an annual monitoring report using a consistent indicator set, since the current practice of revising methodology between publications makes it difficult to assess progress against the carbon budgets.

On governance and evidence, the Secretariat recommends that the Climate Advisory Council be given a statutory right of access to modelling inputs held by line ministries, and that the underlying scenario assumptions be published alongside each carbon budget proposal. Ex post evaluation should be commissioned at the point of instrument design rather than retrofitted, with a defined counterfactual and a sample large enough to support sectoral disaggregation. Finally, the review recommends that the authorities develop a structured just transition strategy for the northern industrial corridor, prepared in consultation with regional authorities and social partners, and supported by dedicated funding within the next budget cycle. The Secretariat stands ready to support implementation through peer review and technical assistance, and proposes a mid-term progress assessment in 2027.